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Write brief answer to the following questions

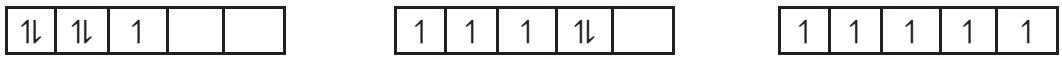
26. Which quantum number reveal information about the shape, energy, orientation and size of orbitals?

27. How many orbitals are possible for n =4?

28. How many radial nodes for 2s, 4p, 5d and 4f orbitals exhibit? How many angular nodes?

29. The stabilisation of a half filled d - orbital is more pronounced than that of the p-orbital why?

30. Consider the following electronic arrangements for the d^5 configuration.



(a) (b) (c)

(i) which of these represents the ground state

(ii) which configuration has the maximum exchange energy.

31. State and explain pauli's exclusion principle.

32. Define orbital? what are the n and l values for $3p_x$ and $4d_{x^2-y^2}$ electron?

33. Explain briefly the time independent schrodinger wave equation?

34. Calculate the uncertainty in position of an electron, if $\Delta v = 0.1\%$ and $\upsilon = 2.2 \times 10^6$ ms^{-1}

35. Determine the values of all the four quantum numbers of the 8th electron in O- atom and 15^{th} electron in Cl atom and the last electron in chromium

36. The quantum mechanical treatment of the hydrogen atom gives the energy value:

$$E_n = \frac{-13.6}{n^2} \text{ ev atom}^{-1}$$

i) use this expression to find ΔE between n = 3 and m=4

ii) Calculate the wavelength corresponding to the above transition.